

<Link> component enables client-side navigation in React Router without page refresh, unlike anchor tags which trigger full page reloads and lose application state—making **<Link>** essential for maintaining a seamless single-page application experience.

Route guards are mechanisms to control access to routes based on conditions like authentication, implemented by creating protected route components that conditionally render content or redirect users using `Navigate` component when authorization fails.

<Switch> component (now **<Routes>** in v6) renders the first matching route exclusively, preventing multiple routes from matching simultaneously—beneficial in scenarios like nested dashboards where you need to display only one specific page at a time.